

José Lopes

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EDUCATION

Instituto Superior Técnico

Lisbon, Portugal

Bachelor of Science in Computer Science and Engineering

Sep 2021 — Jun 2024

- Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Discrete Mathematics, General Software Engineering

Instituto Superior Técnico

Lisbon, Portugal

Master of Science in Computer Science and Engineering

Sep 2024 — Nov 2026 (expected)

- Relevant Coursework: Programming Language Theory, Formal Software Verification, Model Checking, Type Systems Theory, Advanced Programming Topics

PROJECTS & RESEARCH

Master's Thesis - Automatic Test Generation for Node.js Packages

Nov 2025 — Present

- Developing a system that automatically generates test cases for Node.js packages, leveraging a novel type inference mechanism
- Partially funded by a research scholarship from Instituto de Telecomunicações

Type Inference System for Node.js Packages

Mar 2026 — Present

- Building a new type inference mechanism for JavaScript programs using a constraint-based approach. It serves as the core inference engine for the thesis system above, driving automatic test generation

Functional Language Interpreter (github.com/joserlopes/Programming-Languages)

Apr 2025 — Jun 2025

- Implemented an interpreter for a simple functional language as part of the Programming Languages course

EXTRACURRICULAR ACTIVITIES

Resident & Activity Volunteer - Montes Claros University Residence, Lisbon

Sep 2023 — May 2026

- Helped organize and run cultural, social, and academic activities for residents, including welcoming and supporting newly arrived students
- Volunteered on a week-long international service trip to Tangier, Morocco as part of an 8-person group, supporting children at an orphanage and assisting individuals with intellectual disabilities

Youth Monitor/Mentor (Volunteer) - Clube Xénon, Lisbon & Clube Ónix, Braga

Sep 2019 — Jul 2024

- Supervised and mentored groups of school-age students during weekly activities, outings, and multi-day events, safeguarding participants' well-being while maintaining a positive group environment

SKILLS

- **Programming Languages:** Rust, Python, Java, JavaScript, C
- **Technologies:** Git, Linux, Neovim, Docker, Terminal
- **Languages:** Portuguese (native), English (fluent)